

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	17 June 2026
Team ID	
Project Name	VIP-C2-ShopEZ
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through Form Registration through Gmail Registration through LinkedIn
FR-2	User Confirmation	Confirmation via Email Confirmation via OTP
FR-3	Authentication & Login	Login via Email and Password Secure JWT token generation
FR-4	Live Market Explorer	Search real-time stock symbols View open/high/low prices Execute "Buy" and "Sell" virtual transactions
FR-5	User Dashboard (Portfolio)	Initialize new accounts with a \$100,000 balance Display an interactive Recharts asset pie chart Render a responsive stock holdings table

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The platform must feature an intuitive, beginner-friendly interface tailored for novice investors, including visual charting and graceful UI fallbacks (e.g., clean 404 dashboards).
NFR-2	Security	The application must enforce secure JWT-based session authentication and strict role-based access control (RBAC) to protect sensitive routes like the Admin Control Room.
NFR-3	Reliability	The system must handle exceptions robustly, catching database connection errors or compiler parsing issues to prevent hard crashes and maintain platform stability.
NFR-4	Performance	The backend must swiftly process incoming requests and fetch external real-time stock API data without causing latency or blocking the main frontend thread.
NFR-5	Availability	The Node.js server and MongoDB database should be deployed to ensure high uptime, allowing users to practice trading whenever the live markets are active.
NFR-6	Scalability	The stateless MERN architecture must be designed to easily scale, supporting future additions like new asset classes (crypto, forex) or an expanding concurrent user base.